

Define a Problem Statement

ServiceNow Enterprise Implementation: Metro Ticket Booking System

APPLICATION: ServiceNow App Engine / Service Portal	DOCUMENT STAGE: Requirements & Architecture Definition	TARGET AUDIENCE: Business Analysts, ServiceNow Developers
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1. EXECUTIVE OVERVIEW

Core Objective: Replace disjointed, manual, and third-party metro transit ticketing requests with an integrated, enterprise-grade **Metro Ticket Booking Module** hosted on ServiceNow (Service Portal & Now Mobile). This solution optimizes employee transit booking, automates manager/travel desk approvals, and integrates directly with Metro Transit APIs for instant QR code generation.

2. CONSOLIDATED PROBLEM STATEMENT MATRIX

The table below breaks down the primary business challenges into operational gaps, affected stakeholders, and quantifiable organizational impacts.

PROBLEM CATEGORY	CURRENT GAP / CHALLENGE	IMPACTED STAKEHOLDER	BUSINESS IMPACT
Manual Request Handling	Employees rely on email chains, paper vouchers, or external consumer apps to request metro travel.	Commuting Employees, HR / Travel Desk	High administrative overhead, high turnaround times (>24 hrs for ticket issuance).
Expense & Reimbursement Delay	Post-travel manual expense filing leads to duplicate processing, delayed payouts, and lost receipts.	Finance & Accounts, Employees	Increased error rates in auditing, employee dissatisfaction, manual reconciliation burden.
Zero Travel Visibility	No unified reporting platform to track corporate travel patterns, peak routes, or monthly expenditure.	Executive Management, Operations	Inability to negotiate bulk corporate discounts or optimize travel budgets.
Security & Compliance Risk	Unverified manual claims and lack of authorization workflows before ticket purchasing.	Audit & Compliance Teams	Risk of unauthorized corporate travel claims and policy non-compliance.

3. GAP ANALYSIS & SERVICENOW CAPABILITIES MAP

Mapping existing operational bottlenecks to specific ServiceNow platform components and features.

CURRENT STATE GAP	SERVICENOW FEATURE / ARTIFACT	TECHNICAL IMPLEMENTATION	PRIORITY	COMPLEXITY
Fragmented intake forms	Service Catalog (Item)	Custom Catalog Item with dynamic variable sets (Source, Destination, Date).	HIGH	LOW
Manual approval routing	Flow Designer Workflow	Automated approval rules based on ticket cost, user role, and travel distance.	HIGH	LOW
External ticket generation	REST Integration Hub	Outbound REST API call to Metro Authority Gateway to fetch dynamic QR tickets.	HIGH	MEDIUM
Physical / Manual receipt delivery	Notification Engine / Mobile	Automated Push/Email notification delivering embedded QR Code / Wallet pass.	MEDIUM	MEDIUM
Lack of spend analytics	Performance Analytics	Real-time dashboards tracking spend by department, route frequency, and user trends.	LOW	LOW

4. TECHNICAL SPECIFICATIONS & DATA ARCHITECTURE

Proposed Custom Table Structure in ServiceNow to support the Metro Booking System (Scoped Application: x_snc_metro_book).

TABLE NAME [LABEL]	EXTENDS TABLE	KEY CUSTOM FIELDS / COLUMNS	ARTIFACT TYPE
x_snc_metro_ticket_request [Metro Ticket Request]	task	• u_passenger (Reference: sys_user) • u_source_station (Reference) • u_destination_station (Reference) • u_fare_amount (Currency) • u_qr_code_payload (String / Attachment) • u_ticket_status (Choice: Draft, Pending, Approved, Issued, Cancelled)	CUSTOM TABLE

TABLE NAME [LABEL]	EXTENDS TABLE	KEY CUSTOM FIELDS / COLUMNS	ARTIFACT TYPE
x_snc_metro_station [Metro Station Master]	cmdb_ci or None	• u_station_name (String) • u_station_code (String) • u_line_color (Choice: Red, Blue, Green, Yellow) • u_is_active (True/False)	MASTER TABLE
x_snc_metro_fare_matrix [Fare Configuration]	None	• u_from_station (Reference) • u_to_station (Reference) • u_standard_fare (Currency) • u_pass_type (Choice)	CONFIG TABLE

5. END-TO-END EXECUTION LIFECYCLE

1. Intake User selects stations & date on Service Portal / Mobile app; fare auto-calculates.	2. Validation Flow Designer validates eligibility and triggers approval if cost exceeds threshold.	3. Integration IntegrationHub sends REST call to Metro API to generate dynamic ticket payload.	4. Delivery QR Code ticket is emailed/pushed to user; travel expense record is created automatically.
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6. KEY PERFORMANCE INDICATORS (KPIs) & SUCCESS CRITERIA

METRIC / TARGET AREA	BASELINE (MANUAL PROCESS)	TARGET GOAL (SERVICENOW APP)
Ticket Fulfill Time	24 - 48 Hours	Instant (< 30 Seconds)
Manual Effort Hours	15 mins per request (Travel Desk)	0 mins (100% Automated Flow)
Expense Reimbursement Lag	7 - 14 Business Days	Real-time Auto-posting
User Satisfaction (CSAT)	~55% (Frequent complaints)	> 92% Positive Feedback